

Exact Global Optimality for Packing Ten Unequal Circles

The unequal-circle proof project

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Abstract

We consider ten pairwise non-overlapping circles with radii $r_i = \sqrt{i}$, $1 \leq i \leq 10$, inside a circular container. We give a computer-assisted proof that the minimum container radius is an exact value R_{crit} , defined as the unique solution of two explicit angular equations in a rational rectangle. A matching ten-circle realization exists at this value, and a finite global certificate excludes every smaller radius. The lower-bound proof combines a complete radial and angular case tree, a two-inequality angle barrier for the main seven-circle order, and negative-cycle certificates for the four exceptional cases. The complete composition closes all 14,757,888 sector assignments. The proof decisions in the replay use integer and rational arithmetic; the result is a reproducible computer-assisted proof, not an end-to-end theorem-prover formalization. We do not claim that R_{crit} is identical to a separately computed algebraic root R_0 .

1 Introduction

Let $r_i = \sqrt{i}$ for $1 \leq i \leq 10$. The problem is to find the smallest radius R for which there are centers $p_1, \dots, p_{10} \in \mathbb{R}^2$ satisfying

$$\|p_i\| + r_i \leq R \quad (1 \leq i \leq 10), \quad (1)$$

$$\|p_i - p_j\| \geq r_i + r_j \quad (1 \leq i < j \leq 10). \quad (2)$$

Write R^* for this minimum. The minimum is attained: after fixing a bounded interval of candidate radii, all center coordinates lie in a compact set, and the constraints are closed.

This paper proves exact global optimality, rather than only a lower bound for a selected subfamily or a local statement near a proposed configuration. The proof is a composition of geometric inequalities with finite certificates. Its central idea is to turn each possible placement into one of finitely many radial boxes and cyclic-sector assignments. Most assignments close directly by rational radial or angular-cycle bounds. The remaining cases split into two geometric families:

1. a seven-circle core in one fixed cyclic order, ruled out by two angular sums whose monotonicities point in opposite directions as the radius of circle 10 varies;
2. four exceptional assignments, projected to a nine-circle order and ruled out by a finite negative-cycle certificate.

The angle barrier applies to every residual main-order case by a local radius estimate from the non-overlap and containment constraints. Finally, the same two angular equations that force the lower bound also define a configuration at R_{crit} .

The finite checks are substantial, but the logical roles are simple: the tree proves that every possible configuration reaches a checked case; the analytic lemmas prove that each case rule is valid; and the exact witness proves attainability. Section 6 gives the complete case accounting, while Section 8 describes the replay and its trust boundary.

2 Problem and exact statement

2.1 Feasibility and the candidate value

For $\rho > 0$, call ρ feasible if there are centers satisfying (1)–(2) with $R = \rho$. Feasibility is monotone in ρ : a configuration feasible at ρ is feasible at every larger container radius. Thus a certified infeasibility result at one radius is a lower bound for R^* .

We define the exact candidate through the angular quantities introduced in Section 4. Let

$$\begin{aligned} J_t &= [1.0037160860750841, 1.0037160860750846], \\ J_R &= [8.3034681221114890, 8.3034681221114900]. \end{aligned}$$

The number R_{crit} is the R -coordinate of the unique pair $(t_{\text{crit}}, R_{\text{crit}}) \in J_t \times J_R$ satisfying

$$A(t_{\text{crit}}, R_{\text{crit}}) = 2\pi, \quad B(t_{\text{crit}}, R_{\text{crit}}) = 2\pi. \quad (3)$$

The existence and uniqueness of this pair, as well as the strict upper endpoint inequality, are proved by exact rational interval checks in Section 4. In particular,

$$8.303468122111489 \leq R_{\text{crit}} < 8.303468122111490 = U. \quad (4)$$

The equations, rather than a rounded decimal or a polynomial root identification, are the definition of the exact value in this paper.

Theorem 2.1 (Global optimality). *For ten circles of radii $1, \sqrt{2}, \dots, \sqrt{10}$ in a circular container,*

$$\boxed{R^* = R_{\text{crit}}}.$$

More precisely, no packing exists for any $R < R_{\text{crit}}$, and a packing exists at $R = R_{\text{crit}}$.

The exact v27 radial tree covers every configuration with $R \leq U$. Its residual cases are closed by the angle barrier and the nine-circle negative-cycle certificate. Since $R_{\text{crit}} < U$, these facts cover every putative packing below R_{crit} . The matching construction is given in Section 7.

2.2 Exactness and scope

The value in Theorem 2.1 is an exact root-defined real number, not a claim that the finite decimal in (4) is itself the optimum. Nor do we identify it with the distinct algebraic number R_0 appearing in earlier computations. The certified interval is narrow enough to distinguish the exact target used here, and the geometric definition already suffices to state and prove the optimality result.

3 How the global proof is organized

A packing has ten radial coordinates $s_i = \|p_i\|$ and ten center directions. The proof infrastructure first subdivides the allowed radial domain by rational midpoint splits. On each radial box, exact necessary conditions determine which cyclic orders and angular-sector assignments remain possible. These assignments form the finite ledger called the global radial tree.

At a terminal case, one of the following happens:

- interval-safe radial inequalities contradict containment or non-overlap;

- lower bounds on pair angles contradict the 2π angular budget, often through a Farkas multiplier certificate;
- a difference-constraint graph contains a strictly negative directed cycle;
- the assignment has the main seven-circle order and radial range, so the two-sum angle barrier applies;
- the assignment is one of the four alternative masks, so its nine-circle projection is covered by the negative-cycle tree.

Every split is rational and its children cover the parent box exactly. Thus the finite tree is a cover argument, not a sampling of candidate configurations.

The global cap is

$$U = 8.303468122111490.$$

Because the exact root lies strictly below this cap, it is enough for the tree to cover all configurations up to U ; no comparison of floating approximations is used to bridge a gap.

4 The main seven-circle angle barrier

4.1 Radial and angular necessary conditions

Consider the seven-circle core

$$K = \{2, 5, 6, 7, 8, 9, 10\}, \quad W = K \setminus \{10\} = \{2, 5, 6, 7, 8, 9\}.$$

For a center p_i , put $s_i = \|p_i\|$. For any two circles with $r_i + r_j \leq s_i + s_j$ and $|s_i - s_j| < r_i + r_j$, define their necessary smaller central angle by

$$\phi_{ij}(s_i, s_j) = \arccos\left(\frac{s_i^2 + s_j^2 - (r_i + r_j)^2}{2s_i s_j}\right). \quad (5)$$

The law of cosines and non-overlap imply that the directed angular gap from one center direction to another is at least this smaller angle. In a cyclic order, all directed gaps sum to 2π .

The residual main order is

$$(10, 5, 7, 9, 2, 8, 6).$$

For $i \in W$ put

$$a_i(R) = R - r_i, \quad D_i = r_{10} + r_i.$$

When the six circles in W are at the container wall, the angle between circle 10 at radial coordinate $t = s_{10}$ and circle i is

$$\alpha_i(t, R) = \arccos\left(\frac{t^2 + a_i(R)^2 - D_i^2}{2t a_i(R)}\right). \quad (6)$$

For two wall circles $i, j \in W$, their necessary angle is

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{r_i r_j}{a_i(R) a_j(R)}}. \quad (7)$$

Define

$$C(R) = \beta_{79}(R) + \beta_{92}(R) + \beta_{28}(R) + \beta_{86}(R), \quad (8)$$

$$A(t, R) = \alpha_5(t, R) + \beta_{57}(R) + C(R) + \alpha_6(t, R), \quad (9)$$

$$B(t, R) = \alpha_7(t, R) + C(R) + \alpha_6(t, R). \quad (10)$$

For the stated order, the directed loop $10 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10$ has total angle 2π and therefore requires $A_{\text{radial}} \leq 2\pi$. A second loop $10 \rightarrow 7 \rightarrow 9 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10$ skips circle 5 in the angle count, but circle 5 remains present in the packing. It gives the independent necessary inequality $B_{\text{radial}} \leq 2\pi$. These two inequalities are the key: their dependence on t has opposite signs near the critical pair.

4.2 The exact critical pair

The following derivative formulas explain how the candidate is isolated. For $i \in \{5, 6, 7\}$ let

$$b_i = a_i(R)^2 - D_i^2, \quad H_i = 4a_i(R)^2 t^2 - (t^2 + b_i)^2.$$

Where the angles are in $(0, \pi)$,

$$\frac{\partial \alpha_i}{\partial t} = \frac{b_i - t^2}{t\sqrt{H_i}}, \quad \frac{\partial \alpha_i}{\partial R} = -\frac{a_i(R)^2 + D_i^2 - t^2}{a_i(R)\sqrt{H_i}}. \quad (11)$$

For a wall-wall angle,

$$\frac{\partial \beta_{ij}}{\partial R} = -\left(\frac{1}{a_i(R)} + \frac{1}{a_j(R)}\right) \sqrt{\frac{r_i r_j}{a_i(R)a_j(R) - r_i r_j}}. \quad (12)$$

Exact sign checks establish $A_t > 0$ and $B_t < 0$ throughout $t \in I$ when $R \in J_R$. Separate interval checks establish $A_R < 0$ and $B_R < 0$ for $t \in I$ and $\frac{15947}{2000} \leq R \leq 8.30346812212$:

$$A_t > 0, \quad B_t < 0, \quad A_R < 0, \quad B_R < 0. \quad (13)$$

For example, $A_t > 0$ follows from $(\alpha_5)_t > 0$ and $(\alpha_6)_t > 0$. The sign of $B_t = (\alpha_7)_t + (\alpha_6)_t$ requires a comparison of two positive terms; the verifier reduces it to a quadratic polynomial inequality on the squared interval for t . Its leading coefficient is negative and its endpoint lower bounds are positive, which certifies the sign on the full interval.

To prove existence, write

$$F(t, R) = A(t, R) - B(t, R) = \alpha_5(t, R) + \beta_{57}(R) - \alpha_7(t, R), \quad G(t, R) = B(t, R) - 2\pi.$$

The exact checker proves opposite signs for F on the lower and upper t -edges of the rational rectangle, and opposite signs for G on the lower and upper R -edges. These are whole-edge sign checks: interval bounds on the relevant derivatives account for variation along each edge. Since $F_t > 0$, for every R in the rectangle there is exactly one $t(R)$ with $F = 0$. Continuity and the edge signs for G then give an R with $G = 0$ by the intermediate value theorem. This is precisely a solution of $A = B = 2\pi$.

The solution is unique. At fixed R , $A_t > 0$ gives at most one solution to $A = 2\pi$. If two simultaneous solutions had $R_1 < R_2$, then $A(t_1, R_2) < 2\pi$ and $B(t_1, R_2) < 2\pi$, because both decrease with R . Restoring $A = 2\pi$ at R_2 would require increasing t , while restoring $B = 2\pi$ would require decreasing t , since $A_t > 0$ and $B_t < 0$. Both cannot hold at the same t . Thus the pair is unique. The exact endpoint checks further show

$$t_{\text{crit}} \in J_t, \quad 8.303468122111489 \leq R_{\text{crit}} < 8.303468122111490.$$

All angle and derivative signs just used are checked without floating-point decisions. The verifier evaluates rational enclosures for square roots, uses Machin's identity with alternating arctangent series to enclose π , and uses half-angle arctangent expansions with rational remainder bounds for the inverse trigonometric terms.

4.3 Why the radial coordinates can be replaced by wall values

The main order cases do not assume that the circles $i \in W$ are at the wall. This is important: the theorem does not move disks, and it does not assume the older sufficient conditions for physically pushing disks outward.

For $d_{ij} = r_i + r_j$, differentiation of (5) gives

$$\frac{\partial \phi_{ij}}{\partial s_i} = -\frac{s_i^2 - s_j^2 + d_{ij}^2}{2s_i^2 s_j \sin \phi_{ij}}. \quad (14)$$

The case boxes imply, for $t = s_{10} \in I = [0.8588, 1.1265]$ and $\frac{15947}{2000} \leq R \leq U$,

$$r_{10} + r_i - t \leq s_i \leq U - r_i \quad (i \in W).$$

The left inequality follows from $r_{10} + r_i \leq \|p_{10} - p_i\| \leq s_{10} + s_i$; the right one is containment. The actual verifier rounds the square-root endpoints outward to rational bounds before doing its checks.

For most coordinates, (14) has a directly certified negative sign in each angular sum. Three shared coordinates require a combined derivative check because their two adjacent pair contributions can have opposite signs. For $(i, g, b) = (9, 7, 2), (8, 6, 2), (2, 9, 8)$, exact rational bounds give:

coordinate s_i	lower bound for the combined sign expression
s_9	> 3.193809
s_8	> 0.143756
s_2	> 0.589640

The smaller angular sines in the two potentially adverse pairs are bounded below by $3/5$; the remaining terms have positive rational lower bounds. Consequently both full radial sums are non-increasing in every s_i . The domains of all angles are also checked throughout the substitutions.

Starting from any feasible packing in the main order, replace its six radial coordinates s_i one at a time by $R - r_i$ only inside the angle formulas. This is an analytic comparison, not a physical movement. Since each angular sum is non-increasing as any s_i increases, the wall-value sums satisfy

$$A(t, R) \leq A_{\text{radial}} \leq 2\pi, \quad B(t, R) \leq B_{\text{radial}} \leq 2\pi.$$

This is the wall-free angular barrier.

Lemma 4.1 (A radius floor for the main core). *If a feasible main-core packing has $t = s_{10} \leq 1.1265$, then*

$$R > \frac{15947}{2000} = 7.9735.$$

Proof. Non-overlap of circles 9 and 10 gives $s_9 + t \geq r_9 + r_{10} = 3 + \sqrt{10}$. Containment of circle 9 gives $R \geq 3 + s_9$. Since $\sqrt{10} > 3.1$ and $t \leq 1.1265$,

$$R \geq 6 + \sqrt{10} - t \geq 6 + \sqrt{10} - 1.1265 > 6 + 3.1 - 1.1265 = 7.9735.$$

□

4.4 Excluding every radius below the critical value

Suppose a main-order packing occurs in a tree box. Its disk-10 coordinate lies in I , so Lemma 4.1 puts its radius in the certified derivative range. If $R < R_{\text{crit}}$, then by (4.3), it must satisfy $A(t, R) \leq 2\pi$ and $B(t, R) \leq 2\pi$.

If $t \geq t_{\text{crit}}$, then $A_R < 0$ on the interval from R to R_{crit} , while $A_t > 0$ at R_{crit} , so

$$A(t, R) > A(t, R_{\text{crit}}) \geq A(t_{\text{crit}}, R_{\text{crit}}) = 2\pi,$$

a contradiction. If $t \leq t_{\text{crit}}$, then $B_t < 0$ and $B_R < 0$ imply

$$B(t, R) > B(t, R_{\text{crit}}) \geq B(t_{\text{crit}}, R_{\text{crit}}) = 2\pi,$$

also a contradiction. These two cases cover every value of t . Thus any main-order case satisfying the tree's radial hypotheses is impossible below R_{crit} .

5 The four exceptional cases: a negative-cycle proof

The four remaining assignments have a different large-circle skeleton and are not covered by the preceding seven-circle theorem. Their masks are $(5, 16, 40, 2)$ (three assignments) and $(4, 16, 41, 2)$ (one assignment). After projecting away circle 1, their nine-circle order is

$$P = (10, 3, 7, 5, 8, 4, 6, 9, 2).$$

The only ambiguity in the relative placement of circles 4 and 6 gives the second possible order

$$Q = (10, 3, 7, 5, 8, 6, 4, 9, 2).$$

The exact certificate covers both orders; the four assignments in the global tree project to P .

5.1 From radial boxes to angle-difference constraints

Fix either order. For a radial box $s_i \in [\ell_i, u_i]$, containment gives $0 \leq s_i \leq U_+ - r_i$, where

$$U_+ = 8.30346812212 > U.$$

Non-overlap also requires $s_i + s_j \geq r_i + r_j$. This inequality safely tightens the lower endpoints of each box. If any lower endpoint is zero, the angle lower bound involving that coordinate is set to zero, which is conservative.

For positive radial endpoints, the cosine forced by a pair of circles is

$$c_{ij}(s_i, s_j) = \frac{s_i^2 + s_j^2 - (r_i + r_j)^2}{2s_i s_j}.$$

On each radial rectangle the verifier bounds this expression above by checking its four corners. The reduction to corners is valid because, with one coordinate fixed, this rational function has no interior maximum in the other coordinate; any stationary point is a minimum. Square roots of the integer radii are enclosed by rational lower bounds.

For each pair, a nonnegative rational angle tick q_{ij} is stored. It is accepted only after an exact rational Taylor inequality certifies $\cos(q_{ij}) > c_{ij}$ throughout the box. Floating-point inverse cosine is used only when searching for candidate ticks; it has no role in replay. Consequently any feasible angular difference Δ_{ij} in the fixed cyclic order must satisfy

$$q_{ij} \leq \Delta_{ij} \leq 2\pi_+ - q_{ij},$$

where π_+ is a rational upper bound for π obtained from Machin’s identity and alternating arctangent series.

Assign the directed edge $j \rightarrow i$ weight $-q_{ij}$ and the edge $i \rightarrow j$ weight $2\pi_+ - q_{ij}$. The angle differences are variables satisfying the corresponding difference constraints. A directed cycle of strictly negative total weight is impossible: summing its constraints would assert that zero is at most a negative number.

5.2 Finite certificate

The initial radial domain is covered by rational boxes. Each unresolved box is bisected at a rational midpoint; the verifier checks that the children cover the parent and that no box is duplicated or omitted. Every leaf has a recorded negative cycle whose edge weights are recomputed from the radial box and rational angle ticks.

Table 1: Exact replay of the two nine-circle order certificates.

Order	Nodes	Splits	Negative-cycle leaves	Unresolved
P	231	115	116	0
Q	77	38	39	0
Total	308	153	155	0

Thus both projected orders are excluded for every $R \leq U_+$, including the four global-tree assignments because $U < U_+$.

6 Composition of the global finite proof

The compressed certificate `certificates/skeleton-radial-tree-v27.json.gz` is a complete rational-split tree for the radial domain up to U . Its replay checks

2,401 nodes, 1,200 splits, 1,201 leaves.

The associated angular ledgers account for 14,757,888 sector assignments. Before adding the final geometric theorems, the ledgers had closed 14,757,691 assignments and left 197 open. The remaining assignments divide exactly as follows:

- 193 main-order cases satisfy the wall-free barrier hypotheses;
- Four masks project to order P ; the negative-cycle replay closes them.

Their common core order is $(10, 5, 7, 9, 2, 8, 6)$, and each disk-10 radial interval is contained in $[0.8588, 1.1265]$. The masks are $(5, 16, 40, 2)$ in three cases and $(4, 16, 41, 2)$ in one. The composition verifier also reclassifies 145 core-order terminal models and 2 full-order terminal witnesses through the same seven-circle angle barrier. This removes their dependence on the previous local terminal rule. Each of these 147 cases has the same disk-10 interval, so Lemma 4.1 supplies the radius range required by the barrier. After these links are replayed, all 14,757,888 sector assignments are closed and no residual case remains.

Proof of Theorem 2.1. Let $R < R_{\text{crit}}$. Since $R_{\text{crit}} < U$, every putative packing belongs to the certified radial-tree domain. Every sector assignment is closed either by the original exact radial/angular checks, by the main-order barrier, by the alternative nine-circle certificate, or by the independent reclassification of the 147 local terminals. Thus no packing exists below R_{crit} . Section 7 constructs a packing at R_{crit} , so $R^* = R_{\text{crit}}$. $\square$

7 A matching ten-circle packing

At $(t_{\text{crit}}, R_{\text{crit}})$, put circle 10 at radial distance t_{crit} and put each $i \in W$ at its wall radius $a_i(R_{\text{crit}}) = R_{\text{crit}} - r_i$. Use the following polar angles:

$$\begin{aligned}\theta_{10} &= 0, \\ \theta_5 &= \alpha_5, \\ \theta_7 &= \alpha_5 + \beta_{57} = \alpha_7, \\ \theta_9 &= \theta_7 + \beta_{79}, \\ \theta_2 &= \theta_9 + \beta_{92}, \\ \theta_8 &= \theta_2 + \beta_{28}, \\ \theta_6 &= \theta_8 + \beta_{86} = 2\pi - \alpha_6.\end{aligned}$$

The identities $\alpha_5 + \beta_{57} = \alpha_7$ and $\alpha_7 + C + \alpha_6 = 2\pi$ are exactly the two root equations. They make the eight pairs

$$(10, 5), (10, 7), (10, 6), (5, 7), (7, 9), (9, 2), (2, 8), (8, 6)$$

tangent. The remaining 13 pairs within the seven-circle core are strictly separated throughout the certified root rectangle.

Place the three other centers at the following fixed rational coordinates:

$$p_1 = (-3.9798, 6.1148), \quad p_3 = (6.4777, 0.8849), \quad p_4 = (5.5222, -2.7574).$$

(The terminating decimals here denote exact rationals.) The witness replay checks all 37 non-contact pairs among the ten circles and verifies strict container inclusion for circles 1, 3, 4 and circle 10, uniformly over the rational enclosure of $(t_{\text{crit}}, R_{\text{crit}})$. The six circles in W are on the wall, and all remaining inequalities have positive certified slack. Hence this is an actual feasible configuration at the exact root, not merely a rounded numerical picture.

8 Reproducibility and trust boundary

The complete composition is replayed from the repository root with

```
uv run -locked -no-sync python \
  tools/verify_global_optimality_completion.py
```

The first command checks the tree, case linkage, root theorem, witness, alternative-order certificate, local radius implication, and input hashes, then writes a JSON report under **artifacts/**.

All decisive arithmetic in these replayers is integer or rational arithmetic. Square roots are enclosed with integer square-root bounds; the trigonometric inequalities are reduced to rational Taylor bounds and rational enclosures of π . Floating-point routines may help generate candidate split points or angle ticks, but the verifier rechecks the accepted claims using exact inequalities. This separates fast certificate discovery from the smaller checking computation.

The trust boundary is explicit. This is a computer-assisted proof whose soundness depends on the correctness of the Python verifiers, the Python runtime’s integer and **Fraction** semantics, and the certificate inputs whose hashes the composition checker records. The proof is not formalized end-to-end in Lean. In particular, references to old Lean developments or the MPFI generator do not substitute for the exact replay used here.

The repository’s global-optimality completion note and the `proof/global_completion/` directory contain the individual theorem proofs and certificate details. The composition report records SHA-256 hashes for the tree, certificates, and verifier sources so a replay can be tied to the checked inputs.

9 Conclusion and remaining identifications

The finite tree and its geometric terminal theorems give a global lower bound at the exact angle-defined value R_{crit} . The root construction supplies a ten-circle packing at the same value. Therefore the minimum container radius is exactly R_{crit} , as claimed in Theorem 2.1.

The result does not require an algebraic description of the optimum. A separate task would be to prove that R_{crit} equals the previously computed algebraic root R_0 ; overlapping decimal enclosures alone would not prove that identity. The exact coupled-angle definition is already a complete specification of the optimum.